

# UCT Dev Board (STM32F051C8) – Comprehensive Pinout & Hardware Reference

This markdown document provides a detailed pinout mapping, hardware sub-circuit breakdown, and functional description of the **UCT Dev Board** based on the provided KiCad schematic.

## 1. Project Metadata

- Title:** UCT Dev Board
- Institution:** University of Cape Town
- Original Designer:** James Gowans
- Revision Author:** Justin Pead
- Date:** 2025-01-06
- Revision:** v1.06
- EDA Tool:** KiCad E.D.A. 9.0.7

## 2. Comprehensive Pinout Reference Table (Pins 1–48)

The table below describes the function and connections of all 48 pins on the LQFP48 package of the **STM32F051C8** microcontroller.

| Pin | Pin Name      | Hardware Subsystem / Net   | Description & Connections                                                                                                                                                |
|-----|---------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | VBAT          | Power ( 3V3 )              | Device backup domain power supply. Tied directly to the board's main 3V3 rail.                                                                                           |
| 2   | PC13          | External Pin Header ( P1 ) | General-purpose I/O broken out to the main expansion pin header.                                                                                                         |
| 3   | PC14–OSC32_IN | LCD Control ( RS )/ Header | Connected to <b>NMOS Level Shifter 2</b> ( PC14 $\rightarrow$ PC14_S ) to drive the <b>LCD Screen Character Register Select (RS)</b> pin. Also broken out to Header P1 . |

| Pin | Pin Name       | Hardware Subsystem / Net   | Description & Connections                                                                                                                            |
|-----|----------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4   | PC15-OSC32_OUT | LCD Control ( E ) / Header | Connected to <b>NMOS Level Shifter 1</b> ( PC15 $\rightarrow$ PC15_S ) to drive the <b>LCD Screen Enable (E)</b> pin. Also broken out to Header P1 . |
| 5   | PF0-OSC_IN     | Main Clock ( OSC_IN )      | Connected to the 8MHz Main Crystal oscillator circuit ( XTAL ) with a 10pF stability capacitor ( C11 ) to GND.                                       |
| 6   | PF1-OSC_OUT    | Main Clock ( OSC_OUT )     | Connected to the 8MHz Main Crystal oscillator circuit ( XTAL ) with a 10pF stability capacitor ( C12 ) to GND.                                       |
| 7   | NRST           | System Reset               | Active-low reset pin connected to the physical tactile <b>Reset Switch</b> , the onboard ST-Link programming block ( T_NRST ), and Header P1 .       |
| 8   | VSSA           | Analog Ground              | Tied directly to the main system ground ( GND ) plane.                                                                                               |
| 9   | VDDA           | Analog Power Supply        | Analog circuitry power rail. Tied to the 3V3 system voltage line.                                                                                    |
| 10  | PA0            | Switches Block ( SW0 )     | Connected to tactile push-button switch SW0 (pulls to GND when pressed).                                                                             |
| 11  | PA1            | Switches Block ( SW1 )     | Connected to tactile push-button switch SW1 (pulls to GND when pressed).                                                                             |
| 12  | PA2            | Switches Block ( SW2 )     | Connected to tactile push-button switch SW2 (pulls to GND when pressed).                                                                             |
| 13  | PA3            | Switches Block ( SW3 )     | Connected to tactile push-button switch SW3 (pulls to GND when pressed).                                                                             |
| 14  | PA4            | DAC1 Output / HAT          | Digital-to-Analog converter channel 1 output. Labeled DAC1 and routed to Pin 1 of the <b>HAT Connector</b> .                                         |
| 15  | PA5            | External Pin Header ( P1 ) | General-purpose I/O broken out to expansion header P1 .                                                                                              |
| 16  | PA6            | External Pin Header ( P1 ) | General-purpose I/O broken out to expansion header P1 .                                                                                              |
| 17  | PA7            | Loopback Header ( P2 )     | Routed to Loopback Header P2 (Pin 5), designed to couple with PB13 .                                                                                 |
| 18  | PB0            | User LED 0 / Pot Jumper    | Drives individual Red LED D1 via a 150 $\Omega$ resistor. Also routes to the 2x2 Jumper Header to optionally accept analog input from <b>POT0</b> .  |
| 19  | PB1            | User LED 1                 | Drives individual Red LED D2 via a 150 $\Omega$ resistor to GND.                                                                                     |

| Pin | Pin Name | Hardware Subsystem / Net         | Description & Connections                                                                                                                       |
|-----|----------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 20  | PB2      | User LED 2                       | Drives individual Red LED <b>D3</b> via a 150 $\Omega$ resistor to GND.                                                                         |
| 21  | PB10     | Red/Green Status LED             | Labeled <b>PB10</b> . Controls the <b>Red</b> channel of the R/G status LED assembly via a 150 $\Omega$ series resistor ( <b>R12</b> ).         |
| 22  | PB11     | Red/Green Status LED             | Labeled <b>PB11</b> . Controls the <b>Green</b> channel of the R/G status LED assembly via a 150 $\Omega$ series resistor ( <b>R13</b> ).       |
| 23  | VSS      | Digital Ground                   | Primary digital return path connected to the global ground plane ( <b>GND</b> ).                                                                |
| 24  | VDD      | Digital Power Supply             | Main digital core supply input tied to the filtered <b>3V3</b> power rail.                                                                      |
| 25  | PB12     | EEPROM Chip Select ( <b>CS</b> ) | Master-controlled SPI Chip Select line. Hardwired to Pin 1 ( <b>CS</b> ) of the <b>CAT25040LI-G</b> EEPROM IC.                                  |
| 26  | PB13     | EEPROM <b>SO</b> / Loopback      | Connected to Pin 2 ( <b>SO</b> / MISO) of the SPI EEPROM. Also routes to Loopback Jumper Header <b>P2</b> (Pin 6) to connect with <b>PA7</b> .  |
| 27  | PB14     | EEPROM <b>SI</b> / Loopback      | Connected to Pin 5 ( <b>SI</b> / MOSI) of the SPI EEPROM. Also routes to Loopback Jumper Header <b>P2</b> (Pin 4) to connect with <b>PA11</b> . |
| 28  | PB15     | EEPROM <b>SCK</b> / Loopback     | Connected to Pin 6 ( <b>SCK</b> ) of the SPI EEPROM. Also routes to Loopback Jumper Header <b>P2</b> (Pin 2) to connect with <b>PA8</b> .       |
| 29  | PA8      | Loopback Header ( <b>P2</b> )    | General-purpose I/O tied to Loopback Header <b>P2</b> (Pin 1) to form a pair with <b>PB15</b> .                                                 |
| 30  | PA9      | ST-Link UART ( <b>VCP_TX</b> )   | Configured as USART1 Transmit. Interfaces with the ST-Link V2.1 MCU section as the Virtual COM Port TX line ( <b>VCP_TX</b> ).                  |
| 31  | PA10     | ST-Link UART ( <b>VCP_RX</b> )   | Configured as USART1 Receive. Interfaces with the ST-Link V2.1 MCU section as the Virtual COM Port RX line ( <b>VCP_RX</b> ).                   |
| 32  | PA11     | Loopback Header ( <b>P2</b> )    | General-purpose I/O tied to Loopback Header <b>P2</b> (Pin 3) to form a pair with <b>PB14</b> .                                                 |
| 33  | PA12     | LCD Data Bit 6                   | Connected to <b>NMOS Level Shifter 5</b> ( <b>PA12</b> $\rightarrow$ <b>PA12_S</b> ) to drive character <b>LCD Data Line 6 (D6)</b> .           |
| 34  | PA13     | SWD Debug Data ( <b>SWDIO</b> )  | Hardware Serial Wire Debug Data I/O line connected directly to the onboard <b>ST-Link V2.1</b> debugger.                                        |
| 35  | PF6      | <b>HAT Connector</b><br>Pin 5    | Low-pin count GPIO variant routed natively to Pin 5 of the 10-pin expansion HAT connector.                                                      |

| Pin | Pin Name | Hardware Subsystem / Net  | Description & Connections                                                                                                                    |
|-----|----------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 36  | PF7      | HAT Connector Pin 3       | Low-pin count GPIO variant routed natively to Pin 3 of the 10-pin expansion HAT connector.                                                   |
| 37  | PA14     | SWD Debug Clock ( SWCLK ) | Hardware Serial Wire Debug Clock line connected directly to the onboard <b>ST-Link V2.1</b> debugger.                                        |
| 38  | PA15     | LCD Data Bit 7            | Connected to <b>NMOS Level Shifter 6</b> ( PA15 $\rightarrow$ PA15_S ) to drive character <b>LCD Data Line 7 (D7)</b> .                      |
| 39  | PB3      | Byte of LEDs ( D3 )       | Part of the output byte block; drives individual Red LED D3 via a 150 $\Omega$ resistor.                                                     |
| 40  | PB4      | Byte of LEDs ( D4 )       | Part of the output byte block; drives individual Red LED D4 via a 150 $\Omega$ resistor.                                                     |
| 41  | PB5      | Byte of LEDs ( D5 )       | Part of the output byte block; drives individual Red LED D5 via a 150 $\Omega$ resistor.                                                     |
| 42  | PB6      | Byte of LEDs ( D6 )       | Part of the output byte block; drives individual Red LED D6 via a 150 $\Omega$ resistor.                                                     |
| 43  | PB7      | Byte of LEDs ( D7 )       | Part of the output byte block; drives individual Red LED D7 via a 150 $\Omega$ resistor.                                                     |
| 44  | BOOT0    | Boot Mode Configuration   | Boot control pin tied to a 10k $\Omega$ pulldown resistor ( R11 ) to ensure the MCU starts execution from main user Flash memory by default. |
| 45  | PB8      | LCD Data Bit 4            | Connected to <b>NMOS Level Shifter 3</b> ( PB8 $\rightarrow$ PB8_S ) to drive character <b>LCD Data Line 4 (D4)</b> .                        |
| 46  | PB9      | LCD Data Bit 5            | Connected to <b>NMOS Level Shifter 4</b> ( PB9 $\rightarrow$ PB9_S ) to drive character <b>LCD Data Line 5 (D5)</b> .                        |
| 47  | VSS      | Digital Ground            | Core electrical ground return tied to the primary system plane.                                                                              |
| 48  | VDD      | Digital Power Supply      | Main digital core supply input tied to the filtered 3V3 power rail.                                                                          |

### 3. Hardware Subsystems Deep-Dive

#### A. Power Supply & Regulation Block

- Input Sources:** Power is drawn primarily via the standard **USB Connector** (Molex 61729-0010BLF) which provides a raw 5V\_USB rail. An external power terminal block ( JP2 Header 2) is available

for standalone operation.

- **Voltage Regulator:** An `MCP1702-3302E/TO` Low-Dropout (LDO) linear regulator down-converts the raw 5V input down to a stabilized `3V3` system rail.
- **Decoupling Network:** The board integrates an extensive array of filtering capacitors to guarantee clean voltage supply inputs across all frequencies:
  - Bulk Input Filtering: `C1` ( $10\,\mu\text{F}$ ) and `C2` ( $1\,\mu\text{F}$ ).
  - Core Output Smoothing: `C3` ( $5\,\mu\text{F}$ ).
  - High-Frequency Noise Suppression: Parallel network containing capacitors `C4` , `C5` , `C6` , and `C7` (all  $100\,\text{nF}$ ) distributed across supply rails.
- **Visual Indication:** A green **Power LED** ( `D8` ) is connected in series with a  $150\,\Omega$  resistor ( `R10` ) directly across the `3V3` output track to confirm operational voltage status.

## B. User Interface Elements (Buttons & LEDs)

- **The "Byte of LEDs":** A unique design feature of the UCT Dev Board is the provision of an entire continuous 8-bit output byte mapped entirely onto **Port B**.
  - Lower Bits ( `PB0` , `PB1` , `PB2` ) route to individual indicator LEDs ( `D1` , `D2` , `D3` ).
  - Upper Bits ( `PB3` , `PB4` , `PB5` , `PB6` , `PB7` ) drive the remaining 5 LEDs in the cluster assembly.
  - Every individual LED incorporates a  $150\,\Omega$  current-limiting resistor to guarantee uniform brightness.
- **Push-Button Switches:** Four uncommitted tactile input pushbuttons are grouped as the **Switches** block, mapped to pins `PA0` through `PA3` . These interface lines are hardwired to pull down directly to `GND` when depressed.
- **Analog Interface (Pots + Jumper):** An on-board  $1\,\text{k}\Omega$  linear potentiometer ( **POT0** ) serves as a clean adjustable analog voltage source. Using the 2x2 header layout ( `Header 2X2` ), a user can install a physical jumper block to route the variable analog voltage signal directly to `PB0` for ADC conversion testing.

## C. Alphanumeric LCD Character Screen Interface

The board integrates a **WC1602A** character LCD screen module powered by an independent **LCD\_PWR** track. Because standard character screens operate at standard 5V logic thresholds while the STM32F0 operates at 3V3, the design includes **six separate bidirectional N-Channel MOSFET Level Shifters**.

- **Shifter Architecture:** Each of the 6 isolated shifters relies on a discrete NMOS transistor (e.g., **Q2** through **Q6** across schematic sheets) matched with dual  $1.8\text{k}\Omega$  pullup resistors tied between the low-voltage side ( **3V3** ) and the high-voltage side ( **5V** ).
- **Control Configuration:** \* **PC15** shifts to **PC15\_S**  $\rightarrow$  Drives **RS (Register Select)**.
  - **PC14** shifts to **PC14\_S**  $\rightarrow$  Drives **E (Enable Clock)**.
  - The **R/W (Read/Write)** track is permanently pulled low ( **GND** ), locking the module into Write-Only mode to optimize pin constraints.
- **Data Configuration:** To maximize system efficiency, the screen runs in compact **4-bit Nibble Mode** using high-speed data lines 4 to 7:
  - **PB8** shifts to drive **DB4**
  - **PB9** shifts to drive **DB5**
  - **PA12** shifts to drive **DB6**
  - **PA15** shifts to drive **DB7**
- **Contrast Potentiometer:** A dedicated trimmer potentiometer (**POT2**,  $10\text{k}\Omega$ ) interfaces with the LCD contrast pin ( **V0** ) to handle character clarity calibration.

## D. Integrated SPI EEPROM Storage

For non-volatile memory development, the board mounts a **CAT25040LI-G** 4-Kilobit SPI EEPROM IC:

- **Pin 1 ( CS ):** Driven by **PB12** .
- **Pin 2 ( SO ):** Main MISO out, mapped to **PB13** .
- **Pin 5 ( SI ):** Main MOSI in, mapped to **PB14** .
- **Pin 6 ( SCK ):** Bus Clock input, mapped to **PB15** .
- **Loopback Validation Header ( P2 ):** The 2x4 jumper header layout ( **Conn\_02x04\_Odd\_Even** ) provides quick bus manipulation. A programmer can bridge these contacts to link the active EEPROM SPI pins directly to alternative GPIO nodes ( **PA7** , **PA11** , **PA8** ) for diagnostic signal tracing or peripheral separation.

## E. Onboard ST-Link V2.1 Programmer/Debugger

The topmost section features an integrated **ST-Link V2.1** circuitry subsystem that removes the need for external hardware debug tools.

- **SWD Target Connection:** Connects directly to the target via the standard Serial Wire Debug framework: `T_SWCLK` to Pin 37( `PA14` )and `T_SWDIO` to Pin 34( `PA13` ).
- **Virtual COM Port (VCP):** Bridges the microcontroller's internal hardware USART1 lines to the central USB connection, setting up full-duplex serial monitoring over lines `VCP_TX` (Pin 30, `PA9` )and `VCP_RX` (Pin 31, `PA10` ).